Project Milestone #2 - Voyage Planner
Link to App: https://opal.google/app/1BBylBsO1-dg1VhMZTOD8Kx9Dd_W9Hqib

Multi Agent Workflow Design

Overview
The goal of this milestone is to convert the travel-destination app into a multi agent system. Instead of using a single AI step, the app now uses multiple specialized agents that work together in a pipeline. Each agent has a specific role and communicates with the next agent using structured outputs.

The app helps users choose a travel destination based on their preferences such as weather, budget age group, travel type, and personal travel needs.

System Architecture
This system follows a fixed pipeline workflow.

User Input -> Planner Agent -> Executor Agent -> Verifier Agent -> Final Output

Each agent performs a specific task and passes its result to the next agent. This improves the quality of recommendations and makes the system easier to maintain and improve.

Agent Roles
Agent 1: Travel Preference Analyzer(Planner)
This agent receives all user inputs from asking questions. Inputs include: dream travel location, current location, weather preference, budget, age group, travel type, and travel needs.

The planner agent converts these inputs into a structured travel profile. Instead of sending raw text directly to the AI model, this agent organizes the data into categories such as climate preference, budget level, and travel style.

For example:
Preferred climate: Warm, but not hot
Budget: High
Travel style: Leisurely and relaxing
Age group: Young couple
Special needs: Bringing dog along

This structured output is then passed to the tool agent.

Agent 2: Destination Generator (Tool-Using Executor Agent)
This agent uses Google Gemini to generate travel recommendations based on the structured profile created by the planner agent.

The agent generates 3-5 destination suggestions and a brief explanation for each one. Because this agent focuses mainly on generating destinations, it is able to generate more relevant and accurate results than a single step AI agent.

Agent 3: Verifier Agent
The third agent checks whether the generated destinations actually match the user's preferences. If the output does not match the budget, weather, or travel type, the agent improves the recommendations before sending the final result to the user. This agent also improves the explanation by making the results clearer and more personalized.

Communication Between Agents
The agents communicate using a sequential pipeline:

- Agent 1 sends a structured travel profile to Agent 2;
- Agent 2 generates travel destinations;
- Agent 3 verifies and improves the results;
- The final response is shown to the user.

This design ensures that each agent performs a single task, which improves reliability and output quality.

Handling Low-Quality Output
If the generated destinations do not match the user's preferences, the verifier agent will:

- Remove irrelevant destinations
- Regenerate better suggestions
- Improve the explanation

This prevents incorrect or low-quality recommendations from being shown to the user.

Benefits of the Multi-Agent Workflow
The new multi agent design improves the application in several ways:

- More accurate travel recommendations
- Better explanations for users
- Easier debugging and improvement
- Clear separation of responsibilities between agents
- More scalable design for future improvements

Agents Used

Agent 1 - Travel Preference Analyzer

The travel preference analyzer reads the user inputs when they are prompted to answer questions regarding; their dream destination, current location, weather preference, budget, age group attending, travel type and travel needs. Once the user has answered these questions, the travel preference analyzer converts these answers into a structured travel profile and uses it to research multiple destinations that will be later presented to the user.

Agent 2 - Destination Generator (Tool-Using Executor)

The destination generator uses Google Gemini to research and gather potential travel destinations that match up to the users preferences. It generates 3-5 travel destination recommendations so that the user is presented multiple options and can choose from which one appeals to them. These results are generated by matching the budget, weather, travel type, and user age group to the user.

Agent 3 - Verifier Agent

The verifier agent checks whether the recommendations actually match the user inputs. Before the results are presented to the user, the agent double checks their work and ensures it is accurate and useful. If not, the agent redo's the search and generates a better output. The verifier agent improves the explanation for each travel destination and provides details on each result and why it matches up to the user's needs. In addition, it provides quick tips and facts about the destination and why they should book there. In addition, the verifier agent produces the final response that is shown to the user. They present the travel destinations in an organized bulleted list, with details below each potential destination.

Architecture Diagram